LLM FINE-TUNING PROJECT

Fine-Tuning LLM for Text-to-SQL Generation

LoRA fine-tuning on Gretel's synthetic dataset with local MLX inference

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Task & Dataset

Given a **database schema** and a **natural language question**, generate the correct SQL query with explanation.

DATASET
gretelai/synthetic_text_to_sql

LICENSE
Apache 2.0

TOTAL RECORDS
105,851

DOMAINS
100

SQL COMPLEXITY COVERAGE

- Basic SQL
- Aggregation
- Single Join
- Multiple Joins
- Subqueries
- Window Functions
- Set Operations

EACH RECORD CONTAINS

```
// Example record
sql_context: CREATE TABLE ...
sql_prompt: "Total beds per state?"
sql: SELECT state, SUM(beds)...
sql_explanation: "Groups by state..."
sql_complexity: "aggregation"
```

Fine-Tuning Approach

STEP 1		STEP 2		STEP 3		STEP 4		STEP 5
Data Prep	→	LoRA Train	→	Fuse Model	→	Evaluate	→	Gradio Demo
5K train · JSONL		1000 iters · MLX		Merge adapters		50 test samples		Interactive UI

BASE MODEL

Qwen 2.5-3B Instruct

4-bit quantized · MLX format · 3B parameters

METHOD

LoRA (Low-Rank Adaptation)

16 layers · rank adaptation · only 0.216% params trained (6.65M / 3.09B)

INFRASTRUCTURE

MacBook Pro M4

16GB unified memory · Apple MLX framework · ~33 min training time

TRAINING CONFIG

```
learning_rate: 1e-5  |  batch_size: 1  |  iters: 1000  |  lora_layers: 16  |  max_seq_length: 1024  |  val_batches: 25
```

Evaluation Results

EXACT MATCH

+4.0%

26.0% → 30.0%

BLEU SCORE

+12.3%

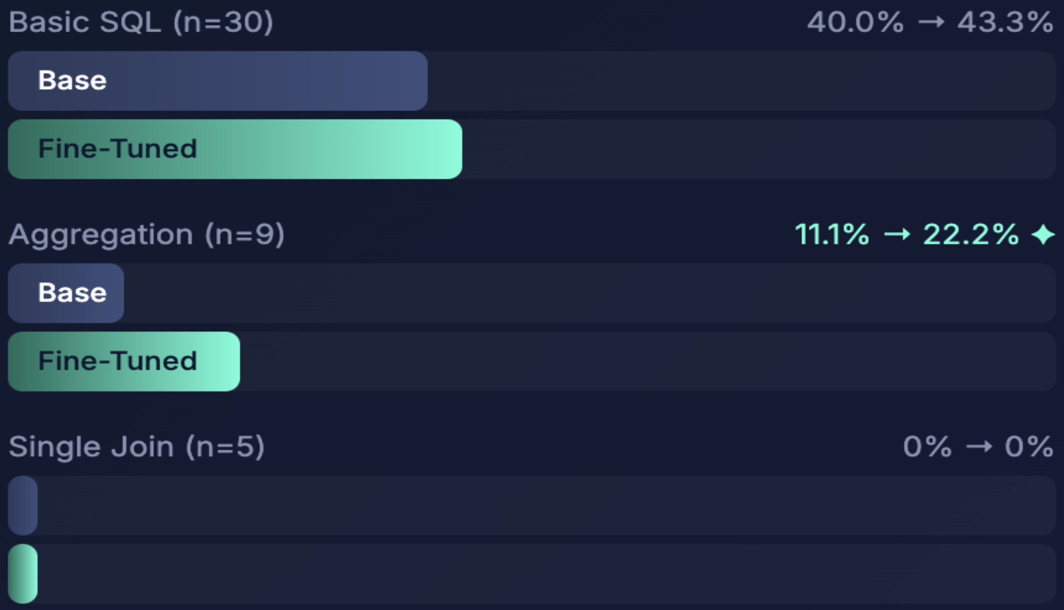
0.504 → 0.566

KEYWORD OVERLAP

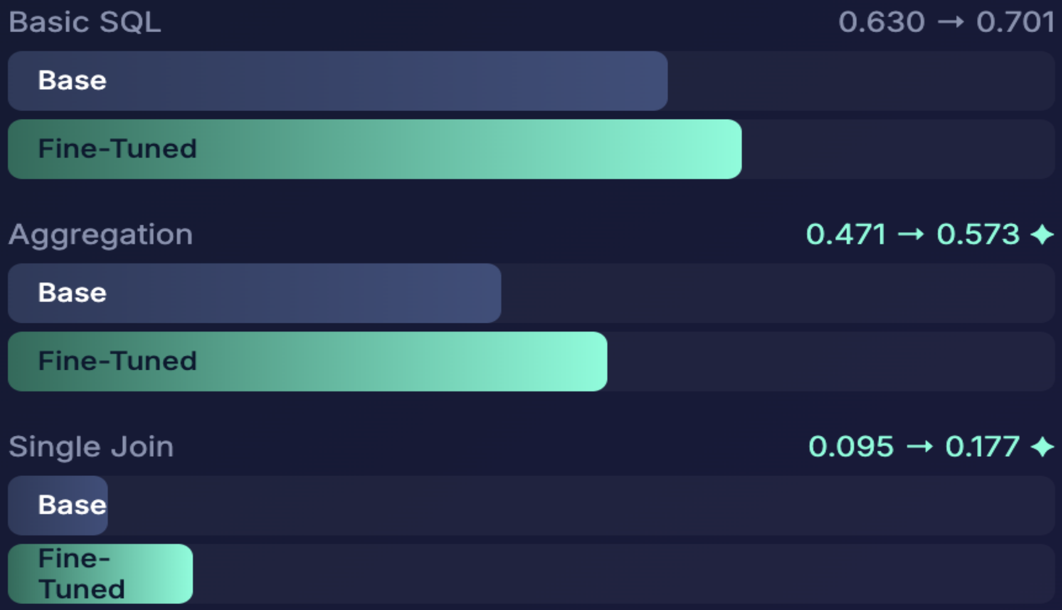
+4.4%

73.0% → 77.4%

EXACT MATCH COMPARISON



BLEU SCORE COMPARISON



Case Study

EXAMPLE 1 · AGGREGATION

```sql
-- Schema
CREATE TABLE media (
  id INT, title VARCHAR(50),
  location VARCHAR(50), genre VARCHAR(50));

-- Question
"How many distinct genres per location?"
```

✗ BASE MODEL

```sql
SELECT T1.location,
  COUNT(DISTINCT T1.genre)
FROM media AS T1
JOIN media AS T2
  ON T1.location = T2.location
-- ✗ Unnecessary self-join
```

GROUND TRUTH

```sql
SELECT location,
  COUNT(DISTINCT genre)
FROM media
GROUP BY location;
```

✓ FINE-TUNED MODEL

```sql
SELECT location,
  COUNT(DISTINCT genre)
FROM media
GROUP BY location;
-- ✓ Exact match!
```

KEY INSIGHT

Fine-tuned model learned to avoid unnecessary joins and produce cleaner, more direct SQL — matching the training data's style and conventions.

Conclusion & Future Work

KEY TAKEAWAYS

- LoRA enables efficient fine-tuning with only **0.216%** of parameters, trainable on a consumer laptop in ~33 minutes
- Consistent improvement across all three metrics: Exact Match, BLEU, and Keyword Overlap
- Strongest gains on **aggregation queries** (2× exact match, +21.6% BLEU)
- Fine-tuned model produces **cleaner SQL** — avoids unnecessary joins and uses correct column names

LIMITATIONS & FUTURE IMPROVEMENTS

- 3B model has limited capacity for complex SQL (subqueries, window functions) — scaling to 7B+ would help
- Only 5,000 training samples used — full 100K dataset could yield stronger results
- Exact match is overly strict — semantically equivalent SQL scored as incorrect
- Future: add execution-based evaluation (compare query results, not just syntax)

TECH STACK

MLX · LoRA · Qwen 2.5-3B · Gradio · Ollama · HuggingFace · Apple M4